STA 445 S26 Assignment 5

2026-03-17

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy('September 13, 2010')
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
mdy('Sept 13, 2010')
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

Does not work since it is not properly abbreviated, to fix this put Sep not Sept.

c. Sep 13, 2010.

```
mdy('Sep 13, 2010')
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```
mdy('S 13, 2010')
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

Same as other problem, does not work since it is not properly abbreviated, to fix this put Sep not S.

e. 07-Dec-1941.

```
dmy('07-Dec-1941')
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```
mdy('1-5-1998')
```

```
## [1] "1998-01-05"
```

This could be MDY or DMY, we will never know if this is referencing Jan or May…

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy('21-5-1998')
```

```
## [1] "1998-05-21"
```

There is no 21st month, 21 has to be day.

h. 2020-May-5 10:30 am

```r
ymd_hm('2020-May-5 10:30 am')
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
#view(OlsonNames())
ymd_hm('2020-May-5 10:30 am', tz='PST8PDT')
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm('2020-May-5 10:30 am', tz = 'America/Puerto_Rico')
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

```r
bday <- mdy_hm('1-06-2001 6:30', tz='US/Arizona')
bday
```

```
## [1] "2001-01-06 06:30:00 MST"
```

```r
twoday <- now(tz='US/Mountain')
twoday
```

```
## [1] "2026-03-17 15:46:30 MDT"
```

a. Calculate the date of your 64th birthday.

```r
bday + years(64)
```

```
## [1] "2065-01-06 06:30:00 MST"
```

b. Calculate your current age (in years).

```r
life <- bday %--% twoday
age <- as.period(life, units = "years")

currentage <- year(age)
currentage
```

```
## [1] 25
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
next_bday <- bday + years(currentage + 1)
next_bday
```

```
## [1] "2027-01-06 06:30:00 MST"
```

d. The number of *days* until your next birthday.

```r
days_until <- as.numeric(as.period(twoday %--% next_bday), "days")
days_until
```

```
## [1] 293.551
```

e. The number of *months* and *days* until your next birthday.

```r
time_until <- as.period(twoday %--% next_bday)
time_until
```

```
## [1] "9m 19d 14H 43M 29.1086890697479S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
phone_call <- mdy_hm('May 8, 2015 3:00 pm', tz='US/Arizona')
phone_call_new <- with_tz(phone_call, tz = "Pacific/Auckland")
phone_call_new
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

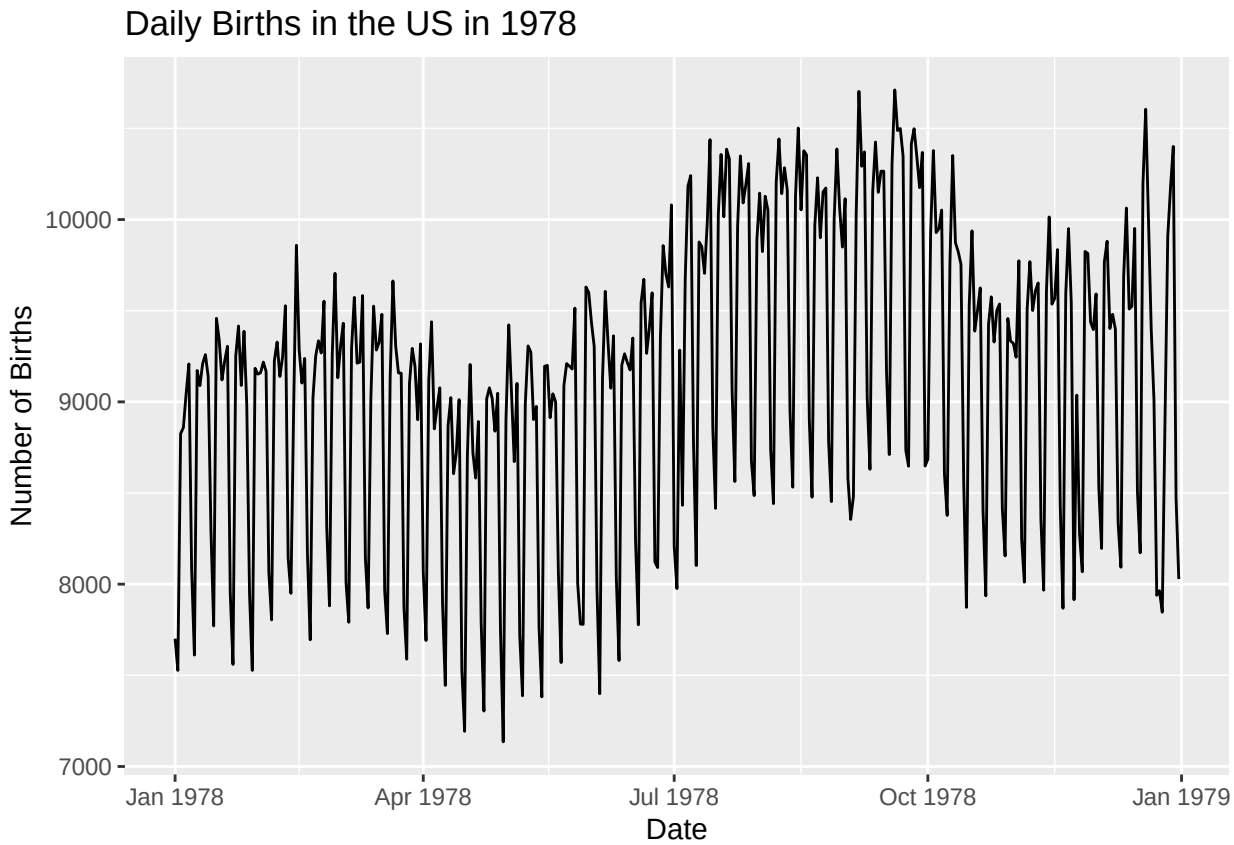
```r
data(Births78)
births_data <- Births78 |>
  select(date, births)
head(births_data)
```

```
##         date births
## 1 1978-01-01   7701
## 2 1978-01-02   7527
## 3 1978-01-03   8825
## 4 1978-01-04   8859
## 5 1978-01-05   9043
## 6 1978-01-06   9208
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```r
ggplot(births_data, aes(x = date, y = births)) +
  geom_line() +
  labs(title = "Daily Births in the US in 1978",
```

```
       x = "Date",
       y = "Number of Births")
```



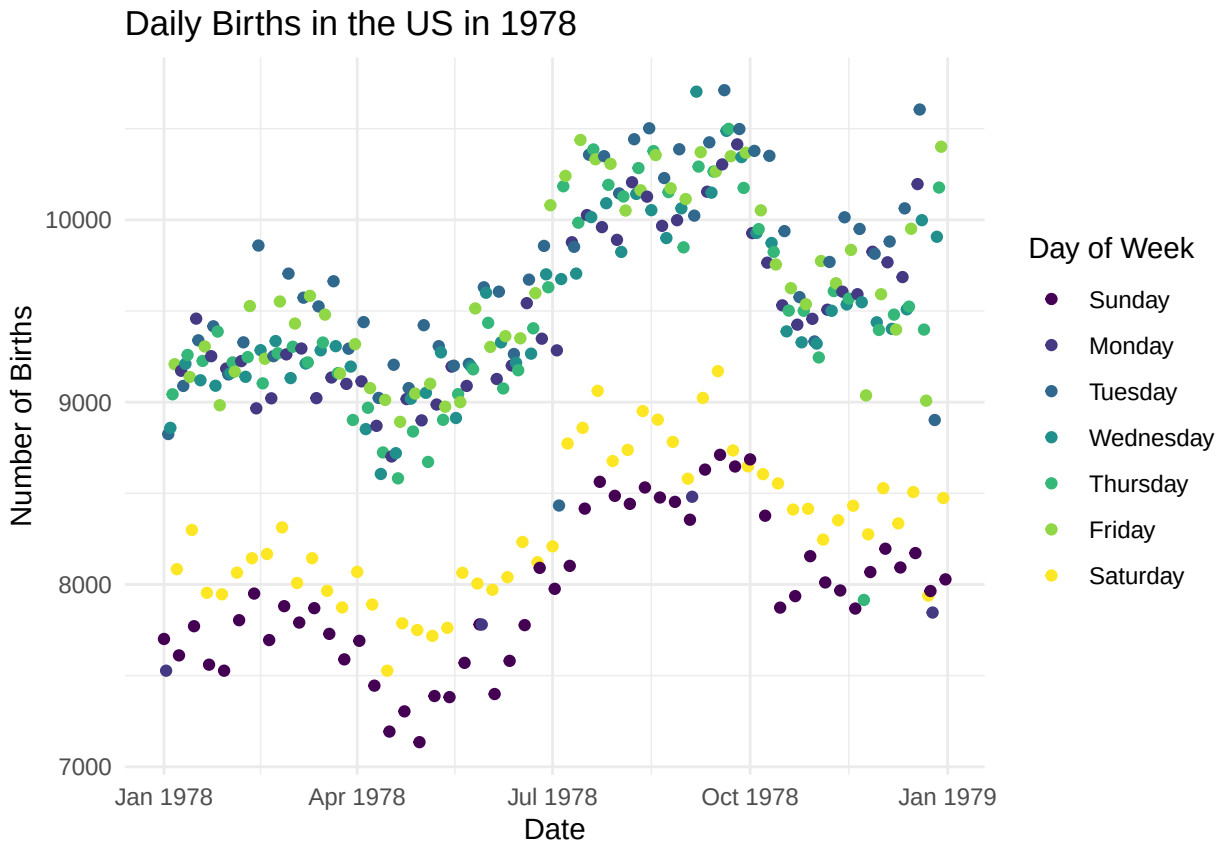
The number of births goes up and down periodically. I assumed this is due to planned births.

c. To test your assumption, we need to figure out what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
births_data <- births_data |>
  mutate(dow = wday(date, label = TRUE, abbr = FALSE))
head(births_data)
```

```
##         date births       dow
## 1 1978-01-01   7701    Sunday
## 2 1978-01-02   7527    Monday
## 3 1978-01-03   8825   Tuesday
## 4 1978-01-04   8859 Wednesday
## 5 1978-01-05   9043  Thursday
## 6 1978-01-06   9208    Friday
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(births_data, aes(x = date, y = births, color = dow)) +
  geom_point() +
  labs(title = "Daily Births in the US in 1978",
       x = "Date",
```

```
      y = "Number of Births",
      color = "Day of Week") +
 theme_minimal()
```